

## **Week .2**

**Course Name : Problem Solving and Testing**

**Course Code : 10211CS227**

**Name : P. Rakesh Kumar**

**Vtu No : Vtu29879**

**1.Build Array from Permutation**

**2. Shuffle the Array**

**3. Find the Highest Altitude**

**4. Java Dequeue**

**5. Java Hashset**

## 1.Build Array from Permutation

### PROGRAM:

```
class Solution {  
    public int[] buildArray(int[] nums) {  
        int[] ans = new int[nums.length];  
  
        for (int i = 0; i < nums.length; i++) {  
            ans[i] = nums[nums[i]];  
        }  
  
        return ans;  
    }  
}
```

### OUTPUT:

☒ Testcase | ☒ Test Result

**Accepted** Runtime: 0 ms

☒ Case 1    ☒ Case 2

Input

nums =  
[0,2,1,5,3,4]

Output

[0,1,2,4,5,3]

Expected

[0,1,2,4,5,3]

## 2. Shuffle the Array

### PROGRAM:

```
class Solution {  
    public int[] shuffle(int[] nums, int n) {  
        int[] ans = new int[2 * n];  
  
        for (int i = 0; i < n; i++) {  
            ans[2 * i] = nums[i];  
            ans[2 * i + 1] = nums[i + n];  
        }  
  
        return ans;  
    }  
}
```

### OUTPUT:

✓ Testcase | > Test Result

**Accepted** Runtime: 0 ms

✓ Case 1    ✓ Case 2    ✓ Case 3

Input

nums =  
[2,5,1,3,4,7]

n =  
3

Output

[2,3,5,4,1,7]

Expected

[2,3,5,4,1,7]

☒ Testcase | [> Test Result](#)

**Accepted** Runtime: 0 ms

☒ Case 1 ☒ **Case 2** ☒ Case 3

Input

nums =  
[1, 2, 3, 4, 4, 3, 2, 1]

n =  
4

Output

[1, 4, 2, 3, 3, 2, 4, 1]

Expected

[1, 4, 2, 3, 3, 2, 4, 1]

☒ Testcase | [> Test Result](#)

**Accepted** Runtime: 0 ms

☒ Case 1 ☒ Case 2 ☒ **Case 3**

Input

nums =  
[1, 1, 2, 2]

n =  
2

Output

[1, 2, 1, 2]

Expected

[1, 2, 1, 2]

### 3. Find the Highest Altitude

#### PROGRAM:

```
class Solution {  
    public int largestAltitude(int[] gain) {  
        int altitude = 0;  
        int highest = 0;  
  
        for (int i = 0; i < gain.length; i++) {  
            altitude += gain[i];  
  
            if (altitude > highest) {  
                highest = altitude;  
            }  
        }  
  
        return highest;  
    }  
}
```

## OUTPUT:

**Accepted** Runtime: 0 ms

☒ Case 1 ☒ Case 2

Input

gain =  
[-5, 1, 5, 0, -7]

Output

1

Expected

1

☒ test case ☒ test result

**Accepted** Runtime: 0 ms

☒ Case 1 ☒ Case 2

Input

gain =  
[-4, -3, -2, -1, 4, 3, 2]

Output

0

Expected

0

## 4. Java Dequeue

### PROGRAM:

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int m = sc.nextInt();

        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

        }

        Deque<Integer> deque = new ArrayDeque<>();

        Map<Integer, Integer> freq = new HashMap<>();

        int maxUnique = 0;

        for (int i = 0; i < n; i++) {

            deque.addLast(arr[i]);
```

```
freq.put(arr[i], freq.getOrDefault(arr[i], 0) + 1);

// Keep window size equal to m
if (deque.size() > m) {
    int removed = deque.removeFirst();

    freq.put(removed, freq.get(removed) - 1);

    if (freq.get(removed) == 0) {
        freq.remove(removed);
    }
}

// Number of unique elements in current window
if (deque.size() == m) {
    maxUnique = Math.max(maxUnique, freq.size());
}
}

System.out.println(maxUnique);
}
}
```

## OUTPUT:

|  |  Console |  I/O |  AI Agent |  |
|--|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--|
|  | 6 3                                                                                       |                                                                                       |                                                                                            |  |
|  | 5 3 2 6 1 8                                                                               |                                                                                       |                                                                                            |  |
|  | 3                                                                                         |                                                                                       |                                                                                            |  |

## 5. Java Hashset

### PROGRAM:

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        Set<String> set = new HashSet<>();

        for (int i = 0; i < n; i++) {

            String first = sc.next();

            String second = sc.next();

            String pair = first + " " + second;

            set.add(pair);

            System.out.println(set.size());

        }

        sc.close();

    }

}
```

## OUTPUT:

|  |  Console |  I/O |  AI Agent |  |
|--|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--|
|  | 4                                                                                         |                                                                                       |                                                                                            |  |
|  | SURYA                                                                                     |                                                                                       |                                                                                            |  |
|  | VAMSHI                                                                                    |                                                                                       |                                                                                            |  |
|  | 1                                                                                         |                                                                                       |                                                                                            |  |
|  | SIDDHU                                                                                    |                                                                                       |                                                                                            |  |
|  | SIVA                                                                                      |                                                                                       |                                                                                            |  |
|  | 2                                                                                         |                                                                                       |                                                                                            |  |